



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA0618 – DESIGN AND ANALYSIS OF ALGORITHM

EXPERIMENT NO: 15

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Program:

```
import math

def distance(p1, p2):

    return math.sqrt((p1[0] - p2[0]) ** 2 +

                     (p1[1] - p2[1]) ** 2)

def closest_pair(points):

    min_distance = float('inf')

    closest = None

    n = len(points)

    for i in range(n - 1):

        for j in range(i + 1, n):

            d = distance(points[i], points[j])

            if d < min_distance:

                min_distance = d

                closest = (points[i], points[j])

    return closest, min_distance

points = [(1, 2), (4, 5), (7, 8), (3, 1)]

pair, min_dist = closest_pair(points)

print("Closest pair =", pair)

print("Distance =", min_dist)
```

Output:

main.py	Visualize	Run ▶	Output
<pre>8 (p1[1] - p2[1]) ** 2) 9 10 11 def closest_pair(points): 12 min_distance = float('inf') 13 closest = None 14 15 n = len(points) 16 17 for i in range(n - 1): 18 for j in range(i + 1, n): 19 d = distance(points[i], points[j]) 20 21 if d < min_distance: 22 min_distance = d 23 closest = (points[i], points[j]) 24 25 return closest, min_distance 26 27 28 # Main program 29 points = [(1, 2), (4, 5), (7, 8), (3, 1)] 30 31 pair, min_dist = closest_pair(points) 32 33 print("Closest pair =", pair) 34 print("Distance =", min_dist)</pre>			<pre>Closest pair = ((1, 2), (3, 1)) Distance = 2.23606797749979</pre>